Northwestern University - Feinberg School of Medicine - Program in Public Health (PPH)

Course Syllabus — PROPOSAL

PUB_HLTH 4XX (number to be assigned) AI and Large Language Models in Healthcare: From Deep Learning to Agentic Systems — 1.0 Unit

- **Term:** Spring Quarter 2027 (10 weeks)
- **Proposed day and time:** Wednesdays, 6:00–9:00 pm (evening section; dates verified against the published 2026–27 Registrar calendar)
- **Location and delivery format:** TBD; the final syllabus will specify the approved in-person, remote, or hybrid format, attendance expectations, recording policy, and contingency arrangements before registration opens
- **Intended concentration:** MPH Data Science (open to other concentrations with instructor permission)

Instructora

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Office hours: by appointment

Teaching Assistant

TBD

Prerequisites

An introductory biostatistics or quantitative methods course (e.g., BIOSTAT 302-0 or equivalent), introduction to statistical learning (BIOSTAT 445-1 or equivalent), and working familiarity with R (e.g., BIOSTAT 306-0 or equivalent experience with data wrangling, functions, packages, and basic modeling). No prior deep learning experience is required; the first two weeks build the necessary foundations. All labs are conducted in R using provided Quarto notebook templates. Students uncertain about their preparation may complete a short pre-course R readiness exercise or contact the instructor.

Course Description

Artificial intelligence, and in particular large language models (LLMs), is rapidly changing how health data are analyzed, how evidence is retrieved and synthesized, and how clinical and public health work gets done. This course gives MPH Data Science students a rigorous, hands-on introduction to modern AI in healthcare, with an emphasis on the two developments most relevant to current practice: deep learning foundations of LLMs and agentic AI systems that use tools, retrieve evidence, and execute multi-step analytic workflows.

The course proceeds in three phases. Weeks 1–4 build conceptual and practical foundations: safe data handling, neural networks, loss functions and optimization, embeddings, the transformer architecture, and how LLMs are trained, aligned, evaluated, and applied to clinical natural language processing tasks such as information extraction, coding, and de-identification. Weeks 5–7 focus on system building: retrieval-augmented generation (RAG) for grounding model outputs in authoritative public health sources, and agentic architectures in which models call tools, query databases, and carry out supervised multi-step analyses. Weeks 8–9 address what makes these systems trustworthy: evaluation and observability, bias and fairness, privacy and regulation, security, and the implementation challenges of deploying AI in real health settings. Week 10 is devoted to project presentations and synthesis.

Throughout, students complete applied R labs on instructor-approved synthetic or license-compatible health data and build toward a final group project: a working, evaluated AI system that addresses a public health problem, accompanied by a written evaluation report, individual contribution statement, and presentation. The course deliberately builds on the R skills students develop elsewhere in the Data Science concentration: the modern R ecosystem for LLM work (ellmer, ragnar, vitals, mcptools, shiny, and shinychat) makes it possible to go from tidy data analysis to grounded, tool-using AI systems without leaving R. The course emphasizes critical appraisal alongside construction. Students should leave able both to build an LLM-based system and to interrogate one built by others.

PPH Competencies

Below are the competencies addressed by this course (final mapping to Data Science concentration competencies to be confirmed with the program office):

- **D1.3** Explain the role of quantitative and qualitative methods and sciences in describing and assessing a population's health
- **D1.6** Explain the critical importance of evidence in advancing public health knowledge
- **D2.1** Apply epidemiological methods to the breadth of settings and situations in public health practice
- **D2.3** Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming and software, as appropriate
- **D2.7** Assess population needs, assets and capacities that affect communities' health
- **D2.19** Communicate audience-appropriate public health content, both in writing and through oral presentation
- **MSB2** Use computer-based statistical analysis package(s) to manage data
- **MSB4** Analyze data employing computer-based statistical analysis package(s)
- **EPI3** Critique the validity of epidemiologic data, findings, and publications by applying knowledge of epidemiologic principles and methods
- **EPI5** Formulate and apply plans for data cleaning and management using statistical analysis software

Learning Objectives

After completion of the course, students should be able to:

1. Explain the core concepts of deep learning — loss functions, gradient-based optimization, embeddings, and attention — and how they give rise to large language model capabilities and

limitations. (D1.3)
2. Apply LLMs to clinical and public health NLP tasks, including information extraction with structured outputs, classification, and de-identification, using R. (D2.3, MSB2, MSB4)
3. Design and implement a retrieval-augmented generation (RAG) pipeline that grounds model outputs in authoritative public health sources with verifiable citations. (D2.3, EPI5)
4. Build a supervised agentic workflow in which a model uses tools (search, database queries, code execution) to complete a multi-step public health analysis with appropriate human oversight. (D2.3, MSB4)
5. Select, classify, and handle health data appropriately for local and hosted AI systems, applying data-use restrictions, privacy safeguards, secrets management, least-privilege tool access, and human approval gates. (D1.6, D2.3, EPI5)
6. Design and execute an evaluation of an LLM-based system, including benchmark construction, baseline comparison, subgroup and error analysis, retrieval and citation metrics, LLM-as-judge methods and their pitfalls, and structured logging/tracing for deployed pipelines. (EPI3, D2.3)
7. Critically appraise published and commercial AI-in-healthcare systems for validity, bias, privacy risk, security, and regulatory posture, and communicate findings to technical and non-technical audiences. (D1.6, EPI3, D2.19)
8. Connect AI system design to epidemiologic purpose, population needs, stakeholder workflow, and implementation science considerations that determine whether such systems improve health in practice. (D2.1, D2.7)

Texts and Readings

There is no required textbook purchase. Readings consist of:

- Keydana S. (2023). *Deep Learning and Scientific Computing with R torch* — free online text; selected chapters for the foundations weeks.
- Peer-reviewed articles (NEJM, Nature, Nature Medicine, npj Digital Medicine, Science, and NeurIPS) as listed and linked in the weekly schedule; all are available through Northwestern Libraries or open access.
- Posit/tidyverse package documentation for the course toolchain: **ellmer** (LLM interfaces), **ragnar** (retrieval-augmented generation), **mcptools** (Model Context Protocol), **vitals** (LLM evaluation), and **shinychat** (chat UIs in Shiny).
- Instructor-provided tutorials and Quarto notebook templates for each lab, plus optional deeper dives listed per week for students who want the underlying technical papers.

Computing Environment

Labs use R ($\geq$ 4.4) in RStudio or Positron with Quarto notebooks. The foundations weeks use **torch for R** to build intuition for gradients, loss functions, and simple networks; the LLM weeks use **ellmer** to call hosted model APIs with typed tool definitions and structured outputs, **ragnar** for retrieval-augmented generation, **mcptools** for agent tool servers, **vitals** for systematic evaluation, and **shiny/shinychat** for lightweight demos.

Course projects use an instructor-provided `renv` lockfile. Submissions record package versions, model/provider identifiers, relevant model settings, retrieval-corpus versions, and access dates.

Cached or mocked responses will be provided for core exercises in the event of an API outage, rate limit, or provider change.

API access for coursework will be provided through instructor-arranged credits; no student payment is required. Only instructor-approved synthetic text or data explicitly authorized for the selected service may be submitted to a hosted API. De-identified patient-derived datasets may still carry data-use, privacy, or contractual restrictions and will be processed locally unless the instructor has documented that external processing is permitted. Students must never submit protected health information, restricted data, credentials, API keys, or other secrets to an external model. Data classification, provenance, permitted processing, and compliance with these rules are graded components of the labs and final project.

Expected weekly effort is three scheduled class hours plus approximately six hours of preparation, practice, and project work outside class. Project-week effort may vary.

Canvas™

The syllabus, readings, notebook templates, and assignment submissions will be hosted on the course's Canvas site, available through Northwestern Canvas for registered students.

Class Assignments, Projects, Quizzes, Tests

Applied labs. In-class R exercises accompany Weeks 1–8. Lab 0 is an ungraded environment and data-safety check; four take-home notebooks are graded (Weeks 2, 4, 5, and 7). Labs are scaffolded with Quarto notebook templates so that students with standard MPH-level R skills can succeed, while extension exercises challenge experienced programmers.

Quizzes. Two short Canvas quizzes (Weeks 3 and 7) check conceptual understanding of deep learning foundations and agentic system design. Quizzes are closed to generative-AI assistance.

Critical appraisal. Each student writes one structured critique, due after Week 9, of a published or commercial AI-in-healthcare system, assessing task framing, data provenance, evaluation validity, subgroup performance, bias, privacy and security risks, regulatory posture, and deployment readiness.

Final project. In groups of 2–3, students design, build, and evaluate an LLM-based system addressing a public health problem (examples: a grounded question-answering assistant over immunization guidance; an agentic surveillance-report summarizer; a structured-extraction pipeline for a registry). The Week 4 proposal identifies the intended user and use, population need, data provenance and permissions, baseline, evaluation metrics, subgroup or failure-mode analyses, security controls, human-oversight plan, and implementation setting. Final deliverables are an in-class presentation, reproducible code, an evaluation report, and an individual contribution/reflection statement. Of the 30 final-system points, 25 are based on the group system and report and 5 on individual contribution evidence and reflection.

Student Assessment

Assignment	Points	Due Dates	Competencies
Quizzes (2; 7.5 points each)	15	Weeks 3, 7	D1.3, D2.3
Applied lab notebooks (4; 7.5 points each)	30	Weeks 2, 4, 5, 7	D1.6, D2.3, MSB2, MSB4, EPI5
Critical appraisal of an AI-in-healthcare study	10	Sunday, May 30, 2027	D1.6, EPI3, D2.19
Final project proposal	5	Week 4	D2.1, D2.7
Final project: working system, evaluation report, code, and individual contribution statement	30	Friday, June 4, 2027	D1.6, D2.1, D2.3, D2.7, MSB4, EPI3, EPI5
Final project presentation	10	Wednesday, June 2, 2027	D2.19

Grading scale

94–100	A	74–76	C
90–93	A−	70–73	C−
87–89	B+	67–69	D+
84–86	B	60–66	D
80–83	B−	0–59	F
77–79	C+		

Generative AI in the Classroom

This is a course about generative AI, and the policy is deliberately open with disclosure: students are encouraged to use generative AI tools throughout the course, including for coursework. Any use of GAI must be accompanied by a disclosure at the end of the assignment describing (1) what GAI was used for, (2) the tool, provider, model/version when available, and date of use, (3) the portions of the work it influenced, (4) a sanitized summary of the prompts or workflow, and (5) how the output was checked. Disclosures must never reproduce protected health information, restricted data, credentials, API keys, or other secrets. Quizzes are closed to GAI assistance. Students remain responsible for factual accuracy, citations, code behavior, data handling, and compliance with assignment requirements. Disclosures are graded for thoughtfulness, not penalized for use.

Course Evaluation

The University administers web-based evaluations to students for each course near the end of the quarter. Completion of both the course and faculty evaluation components is expected. Standard PPH course evaluation language applies.

Northwestern University Syllabus Standards

Northwestern University courses follow the Northwestern University Syllabus Standards. Students are responsible for familiarizing themselves with this information. Standard PPH language on Academic Integrity, Turnitin, Requesting Accommodations (AccessibleNU), Use of Appropriate Names and Pronouns, Diversity Equity and Inclusion, Prohibition of Recording, and Support for Wellness and Mental Health applies as in other PPH syllabi and will be included in the final syllabus.

Class Schedule

Week 1 — 3/31/2027

From regression to deep learning and responsible setup: neural networks, loss functions, and optimization; where AI fits in public health and healthcare; data classification, approved computing environments, API data-retention questions, secrets management, and course safety rules.

Readings / class prep: - Rajkomar A, Dean J, Kohane I. Machine Learning in Medicine. *N Engl J Med.* 2019;380(14):1347–1358. - Keydana, *Deep Learning and Scientific Computing with R torch*: selections on tensors, autograd, and a first neural network. - Optional: Sculley D, et al. Hidden Technical Debt in Machine Learning Systems. *NeurIPS.* 2015.

Due: Lab 0 (ungraded): R/Quarto environment, package, API-access, and data-safety check

Week 2 — 4/7/2027

Learning representations: embeddings, tokenization, and the transformer architecture. Attention, context windows, and what "next-token prediction" does and does not imply.

Readings / class prep: - Keydana, R torch text: selected sections. - 3Blue1Brown. *Attention in transformers, step-by-step | Deep Learning Chapter 6* (video). - Alammar J. *The Illustrated Transformer*. - Optional: Vaswani A, et al. Attention Is All You Need. *NeurIPS.* 2017.

Due: Lab 1: Clinical text embeddings in R (graded)

Week 3 — 4/14/2027

Large language models I: pretraining, instruction tuning, and alignment; capabilities, failure modes, and hallucination; prompting as a programming interface. Introduction to intended-use statements, baselines, evaluation sets, and metrics so that project teams can plan evaluation before building.

Readings / class prep: - Thirunavukarasu AJ, et al. Large language models in medicine. *Nat Med.* 2023;29(8):1930–1940. - Singhal K, et al. Large language models encode clinical knowledge. *Nature.* 2023;620(7972):172–180.

Due: Quiz 1

Week 4 — 4/21/2027

Large language models II: clinical NLP in practice. Information extraction, structured outputs, and selected examples from coding and de-identification; adapting models through prompting, retrieval,

or fine-tuning. De-identification output is treated as a prediction requiring evaluation and review, not as proof that data are safe to disclose.

Readings / class prep: - Wornow M, et al. The shaky foundations of large language models and foundation models for electronic health records. *npj Digit Med.* 2023;6:135. - Instructor-provided tutorial: structured extraction with ellmer type specifications (`type_object`/`type_array`) for schema-constrained outputs.

Due: Lab 2: Structured extraction from synthetic clinical notes (graded); **final project proposal due**, including intended use, data permissions, baseline, evaluation plan, risk controls, human oversight, and implementation setting

Week 5 — 4/28/2027

Retrieval-augmented generation (RAG): chunking, embedding stores, vector and text retrieval, grounding, and citation. Retrieval evaluation, citation correctness, corpus versioning, and indirect prompt-injection risks.

Readings / class prep: - Lewis P, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS.* 2020. - Instructor-provided tutorial: building a RAG pipeline with ragnar over a versioned public-health guidance corpus.

Due: Lab 3: RAG over an instructor-provided, versioned CDC/WHO guidance corpus, including retrieval and citation evaluation (graded)

Week 6 — 5/5/2027

Agentic AI I: tool use and function calling, the Model Context Protocol (MCP), and orchestration patterns; workflows versus agents. Threat modeling, prompt injection, least-privilege tools, secret isolation, code-execution sandboxing, audit logs, and approval gates.

Readings / class prep: - Anthropic. *Building Effective Agents* (2024). - ellmer and mcptools documentation: registering R functions as tools, tool schemas, MCP servers and clients, and agent loops in R.

Due: Critical-appraisal topic and source approval

Week 7 — 5/12/2027

Agentic AI II: multi-step analytic workflows, human-in-the-loop design, guardrails, trajectory review, and failure containment. Agents for data analysis and literature synthesis in public health.

Readings / class prep: - Instructor-provided tutorial: a constrained agentic data-analysis harness in R — the model queries an approved database and runs allowlisted tidyverse analyses via registered tools. - Optional: selected readings on multi-agent systems and evaluation of agent trajectories.

Due: Lab 4: Build and test a constrained tool-using agent in R (graded); Quiz 2

Week 8 — 5/19/2027

Evaluation and observability: benchmark design, baseline comparison, retrieval and citation metrics, subgroup analysis, LLM-as-judge and its pitfalls, human rating and inter-rater agreement, structured logs and traces, error analysis, cost/latency, and monitoring deployed systems.

Readings / class prep: - Zheng L, et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS.* 2023. - Instructor-provided tutorial: systematic evaluation with vitals (datasets, solvers, and scorers) plus separate structured logging and trace records for an LLM pipeline in R.

Due: Project working session: bring a runnable baseline, draft evaluation set, risk register, and initial error analysis

Week 9 — 5/26/2027

Responsible AI and implementation in healthcare: bias and fairness, privacy and the scope and limits of HIPAA, consumer-health data protections, FDA resources on AI-enabled medical devices, ASTP/ONC HTI-1 algorithm transparency, stakeholder workflow, and implementation science. Guest lecture (TBD).

Readings / class prep: - Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science.* 2019;366(6464):447–453. - ASTP/ONC. HTI-1 Final Rule materials on algorithm transparency. - FDA. Artificial Intelligence-Enabled Medical Devices.

Due: Project working session; **critical appraisal due Sunday, May 30**

Week 10 — 6/2/2027

Final project presentations and course synthesis: what changes for the public health workforce when analysis, retrieval, and drafting are agentic.

Readings / class prep: No new readings.

Due: Presentations in class; final report, reproducible code, and individual contribution statement due Friday, June 4

Additional Resources

Books and long-form - Keydana S. *Deep Learning and Scientific Computing with R torch* - 3Blue1Brown Neural Networks series (visual intuition for backpropagation and transformers)

Online resources - Model documentation: Anthropic, OpenAI, and Google AI for Developers - Posit AI documentation: ellmer, ragnar, vitals, mcptools, and shinychat - Implementation-science resources: CFIR and RE-AIM (see PUB_HLTH 351)

Podcasts and current-events resources - NEJM AI Grand Rounds - The Gradient - Latent Space

Links and package availability were last checked August 3, 2026. The instructor will recheck all assigned links, package APIs, and model-provider requirements before the Canvas site opens for Spring 2027.